

Shamitha Gaddam

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Education

University of Pennsylvania, Philadelphia, PA, USA

Aug 2026 – May 2028

Masters in Computer and Information Science

Relevant Coursework: Applied Machine Learning, Big Data Analytics, Natural Language Processing

BNM Institute of Technology, Bangalore, India

Dec 2021 – Aug 2025

Bachelor of Engineering in Artificial Intelligence and Machine Learning

GPA: 3.9/4

Professional Experience

Puma, AI Engineer - Bangalore, India

Oct 2025 – June 2026

- Developed an AI-powered HSN classification platform for apparel, footwear, and accessories using LLMs, rule-based validation, and enterprise APIs, achieving **98% classification accuracy** and automatically classifying 6,000+ retail records in under 7 minutes through a unified BOM, PESA, and merchandising data pipeline.
- Built a multi-agent AI assistant providing conversational access to product, logistics, and merchandising data, enabling business teams to retrieve information and insights across enterprise systems through natural language interactions.
- Designed and deployed AI-driven freight invoice automation workflows for PDFs, EDI, and spreadsheets, achieving **95% processing accuracy** through OCR- and LLM-based extraction, shipment matching, rate-card validation, invoice reconciliation, and data standardization across multiple logistics carriers.

42Gears Mobility Systems, SDE I - Bangalore, India

Jan 2025 – Sept 2025

- Automated SureMDM Agent workflows for macOS package management, eliminating manual intervention and improving deployment efficiency across managed devices.
- Built an AI-powered code review agent using GitHub Copilot and LLM-based workflows to automate merge request analysis, code validation, and review recommendations.
- Developed and enhanced macOS-specific features within the SureMDM Agent, improving platform integration and device management capabilities.

Patents

- **AI-Driven Chemical Compound Analyzer** (*Granted*, App No: 413050-001), Jul 2024
- **AI-Enabled IoT Solar Water Purification System** (*Published*, App No: 202441052962), Jul 2024
- **AI-Driven IoT Soil Nutrient Monitoring System** (*Published*, App No: 202441052961), Jul 2024
- **AI-Powered Smart Refrigerator Inventory Management System** (*Published*, App No: 202441052964), Jul 2024

Publications

- **A Survey on Graph Neural Networks and its Applications** – *Springer SN Computer Science*, Dec 2024
Surveyed GNN architectures and applications across computer vision, NLP, and predictive analytics.
- **Modeling and Performance Analysis of PEM Fuel Cells Using Machine Learning** *IEEE IITCEE*, Jan 2025
Developed ML regression models achieving 94.7% accuracy in predicting fuel cell performance metrics.
- **Detection of Potability of Drinking Water Using Machine Learning** *IEEE ICCAMS*, Jul 2025
Built supervised learning pipelines using SVM, XGBoost, and Random Forest for water quality classification.

Projects

Automatic Metadata Extraction for Text-to-SQL

- Built a text-to-SQL pipeline inspired by test-time compute scaling to enable multi-step reasoning over relational database schemas, including handling joins and ambiguous column mappings.

Pulse — Embodied Conversational AI

- Built a real-time conversational AI system that transforms voice interactions into a living, adaptive visual interface, exploring the future of human–AI communication.

AI-Based Protein Crystallization Detection System

- Built an AI-powered protein crystallization detection pipeline using deep learning and Generative AI, reducing manual inspection effort and radiation exposure while increasing classification accuracy from 85

Skills

Coding Languages: Python, HTML, CSS, Swift, R, SQL

Databases: SQLite, MySQL, MongoDB, PostgreSQL

Frameworks & Libraries: PyTorch, Huggingface, Langchain, Matplotlib, NLTK, Numpy, TensorFlow, Transformers, Scikit-Learn, OpenCV, Terraform, Spacy, FastAPI, REST APIs

Technologies: Git, Azure, AWS, GCP, Docker, Linux, Hadoop, API Gateway, CI/CD, Microservices, WebSockets, Xcode, Figma, Azure OpenAI, Kubernetes, Prompt Engineering, RAGs, GANs